

Team Structure (5 Members)

Member 1

ESP32 & Sensors Engineer

Responsibilities

- Build ESP32 circuit in Wokwi
- Connect all sensors
- Connect relay
- Write MicroPython code
- Publish sensor readings
- Subscribe to control topic
- Test hardware

Deliverables

- Wokwi Project
- ESP32 Code
- MQTT Topics

MQTT Topics

Publish

building/temperature

building/humidity

building/light

building/motion

building/smoke

Subscribe

building/light/control

building/fan/control

Hands project to

➡ Member 2

Member 2

MQTT & Communication Engineer

Responsibilities

- Design MQTT architecture
- Create Topics
- Test Publish/Subscribe
- Connect ESP32 with HiveMQ
- Verify communication
- Document topic structure

Deliverables

- MQTT Topic List
- Communication Tested

Receives

ESP32 code

Delivers

Working MQTT communication

➡ Member 3

Member 3

Node-RED Backend Engineer

Responsibilities

Create Node-RED Flow

Subscribe to MQTT

Parse messages

Store data

Read commands

Publish commands

Create automation rules

Examples

If

Temperature >30



Publish

building/fan/control

ON

If

Light<100

AND

Motion=1



Publish

building/light/control

ON

Store data



MySQL

Receives

MQTT Topics

Delivers

Working Flow

➡ Member 4

Member 4

Database Engineer

Responsibilities

Design Database

Create Tables

Store sensor readings

Create SQL Queries

Generate Reports

Database

Sensors

id

sensor_name

value

timestamp

Automation Logs

id

device

action

time

Device Status

device

status

updated_at

Receives

Node RED

Delivers

Database

➡ Member 5

Member 5

Dashboard & Presentation Engineer

Responsibilities

Create Dashboard

Charts

Cards

Device Status

History

Remote Buttons

Presentation

Documentation

Dashboard

Temperature

Humidity

Smoke

Motion

Light

Relay Status

Fan Status

Buttons

Light ON

Light OFF

Fan ON

Fan OFF

Charts

Daily

Weekly

Live

Receives

Database

Node RED

Creates

Presentation

Demo

Final Report

Parallel Work Plan

Day 1

Member 1

ESP32

Sensors

Relay

MQTT

Member 2

MQTT Topics

Communication

HiveMQ

Testing

Member 3

Create Node RED Flow

MQTT Nodes

Functions

Member 4

Database

MySQL

Tables

Queries

Member 5

Dashboard Design

Presentation Template

Architecture Diagram

End of Day 1

Everything should be connected.

ESP32

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MQTT



Node RED

Day 2

Member 3



Connect



MySQL



Dashboard

Member 1

Tests sensors

Member 2

Tests communication

Member 4

Checks Database

Member 5

Design Dashboard

Day 3

Testing

Fix Bugs

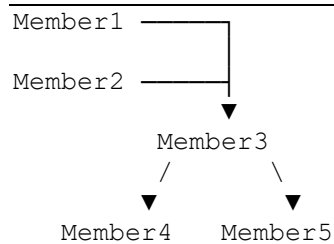
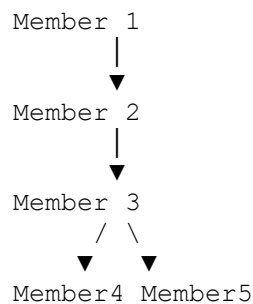
Presentation

Video

Documentation

Demo

Dependency Flow



يعني:

- MQTT و 2 يخلصوا الاتصال والد Member 1.
- تبقى جاهزة Topics يبدأ بمجرد ما الد Member 3.
- Member 3 و 5 يقدروا يشتغلوا بالتوازي مع Member 4.

Suggested Sensors (Simple & Available)

- Temperature (DHT22)
- Humidity (DHT22)

- LDR (Light Sensor)
 - PIR Motion Sensor
 - Gas/Smoke Sensor (MQ-2)
 - Relay (Light/Fan Control)
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Features (Enough for 3 Days)

Monitoring

- Live temperature
- Live humidity
- Light level
- Motion detection
- Smoke/Gas detection

Automation

- Turn light ON when motion is detected in a dark room.
- Turn fan ON automatically when temperature exceeds a threshold.
- Trigger an alarm when smoke/gas is detected.

Remote Control

- Turn Light ON/OFF
- Turn Fan ON/OFF

Data Logging

- Save all readings in MySQL with timestamps.
- Record every automation event.

Dashboard

- Live sensor cards.
- Device status indicators.
- Historical charts.
- Manual control buttons.